

Section 2.3: Sensors

In the last section we looked at how data can be input into a computer system (both manually and directly).

In this section we will discuss how sensors can be used to automatically detect changes in the environment around and then directly feed this data into a computer for processing.

Key Concepts of this section:

- # Know the **definition** of the term '**sensor**'.
- # Understand the **different types of sensors** and what **uses** they perform.
- # Be able to **discuss advantages and disadvantages** of sensors.

Sensors (what are they?)

Definition: "A device which automatically inputs data into a computer system, where the data is constantly changing and can be measured."



Key Words:

Data, Automatic, Measured, Analogue, Digital, Analogue to Digital Converter, ADC

Overview of Sensors:

- # Used to **detect data** in the **surrounding environment** that **constantly changes**. For example:
 - **Temperature** - measuring heat in a room
 - **Light** - setting off a burglar alarm
 - **Humidity** - detecting when plants become too dry.
- # This information is '**physical**' and '**analogue**'.

Examples:



Thermometers are examples of temperature sensors.

Analogue data is **constantly changing** and it changes **very smoothly**.

For example: Outside temperature changes all of the time but it does so very gradually and smoothly. It doesn't suddenly jump from 10°C to 20°C.

Computers don't understand analogue data and therefore it needs to be **converted to digital** (e.g. 1's and 0's).

The computer can then read the data.

A special device called an **Analogue to Digital Converter (ADC)** is used in order to achieve this.



Infrared burglar alarms are examples of light sensors.

Examples of Sensors

The table below shows you some examples of sensors:



Key Words:

Temperature, Pressure, Light, Sound, Humidity, pH, Monitoring, Data Collection

Temperature



Pressure



Light



REMEMBER!

Sensors are used to **AUTOMATICALLY** get data into a computer system.

NOTE:

We will look at each sensor in greater detail further down this page.

Sound

Humidity/Moisture

pH



Uses of Sensors

- # Sensors are used in **monitoring** and **control applications**.
- # When monitoring, the data is sent directly to some sort of a computer and is then **processed and used**.

For example: A burglar climbs through a window and walks through an alarm's light sensor. The computer inside the alarm will pick up on the broken light signal and use this information to sound the alarm

- # Sensors can also be used for a wide range of **data collection** and **everyday applications**.

For example:

- Taking measurements in scientific experiments
- Measuring pollution in the atmosphere
- Sensing the temperature in a room and using this to either turn a heating system on or off
- Automatic doors open themselves when they detect a person nearby.

- # The table below specifies the uses of different types of sensors:

Type of Sensor:	Used For:
Temperature	Automatic washing machines, Central heating systems, Automatic greenhouses, Cookers.

Examples:



Sensors in burglar alarms can automatically detect intruders.



Automatic doors open when people are detected nearby.

Pressure	Burglar alarm systems, Washing machines, Robotics, Environmental monitoring.
Light	Automatic greenhouses, Automatic doors, Burglar alarm systems, Street lighting control.
Sound	Burglar alarm systems, Voice recognition systems (like using a voice command to start a car).
Humidity/Moisture	Automatic greenhouses, Environmental monitoring, Agriculture, Factories that manufacture items that can be ruined by too much air moisture (microchips etc).
pH	Automatic greenhouses, Chemical manufacturing, Environmental monitoring, Aquariums (wrong pH can kill fish).



A variety of sensors are used on automatic greenhouses to keep plants in perfect conditions.

Advantages/Disadvantages of Sensors

The table below highlights some of the main advantages and disadvantages of using sensors:

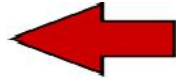
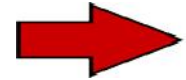
Advantages	Disadvantages
Readings taken using sensors are more accurate than those taken by a human.	Faulty sensors can give incorrect results . For example: if the sensors on a car that help with reverse parking become dirty they may not identify an obstacle and cause you to crash into it.
Sensors can monitor information constantly (They don't get tired like humans can).	
Sensors can respond to information immediately (Burglar alarms goes off as soon as an intruder is detected).	
No need for humans to operate the sensor. This can be useful if gathering the information is hazardous like in detecting radiation levels for example.	



pH sensors are used to monitor the acidity and alkalinity level of aquariums (wrong levels can kill fish).



Malfunctioning sensors used on cars to assist in 'reverse parking' can cause crashes.

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